

Topic:-

Use your imagination and make models of the Milky Way and the Solar System.

Introduction:-

The universe is a vast, mysterious, and awe-inspiring expanse filled with billions of shining celestial bodies. When we look up at the clear night sky away from city lights, we see thousands of twinkling stars. In science, we learn that stars do not exist alone in isolation; they are gathered into gigantic cosmic islands called **Galaxies**, which contain gas clouds, cosmic dust, and hundreds of billions of stars bound together by mutual gravity.

Our cosmic home is called the **Milky Way Galaxy** (known as *Mandakini* or *Akashganga* in Marathi and Sanskrit). It is a gigantic barred spiral galaxy that looks like a glowing pinwheel or a band of spilled milk across the dark celestial sphere. Located inside one of the outer spiral arms of this enormous galaxy lies our own **Solar System**, consisting of our burning Sun at the center and eight planets revolving around it. In this project, we used our imagination, creativity, and simple craft materials to build 3D physical models of the Milky Way and the Solar System.

❖ *Astronomical Scale of the Milky Way Galaxy:-*

- **Diameter:** Spans an incredible 1,00,000 light-years across space!
- **Star Population:** Contains between 100 to 400 billion stars, including our Sun.
- **Our Position:** Our Solar System is located in the Orion Spiral Arm, about 27,000 light-years from the center.

Aim & Craft Materials Required:-

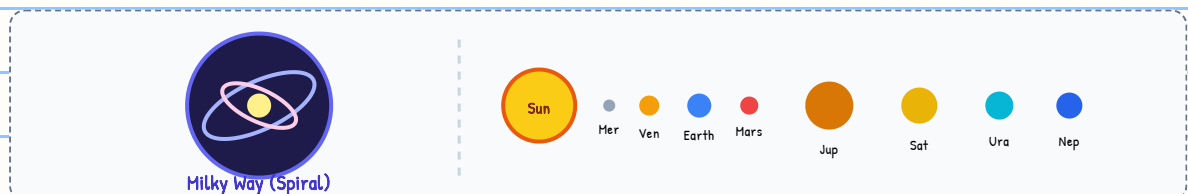
Aim: To design, construct, and present creative 3D models of the spiral Milky Way galaxy and the Solar System using affordable, recycled art and craft materials.

❖ *Objectives of the Project:-*

- To physically represent the barred spiral structure of the Milky Way galaxy.
- To build proportionate models of the Sun and eight planets in their correct orbital sequence.
- To explain relative planetary sizes, distances, and orbits to our classmates.

❖ *Materials Required:-*

- Black cardboard sheet / open shoebox (outer space background), fluffy white cotton wool.
- Papier-mâché dough (newspaper + wheat flour paste) or modeling clay for spherical planets.
- Silver and golden glitter, acrylic poster paints, thin metal wire, wooden skewers, thread, scissors.



Step-by-Step Model Construction:-

We crafted our astronomical models through two distinct creative phases:

- **Phase 1: Building the Spiral Milky Way Model:**

- Took a large circular black cardboard base representing dark outer space.
- Painted a dense golden-yellow glowing sphere in the center representing the galactic nucleus bulge (which harbors a supermassive black hole).
- Arranged soft, fluffy white cotton wool in four curved spiral arms radiating outward.
- Sprinkled silver and blue glitter over the arms to represent billions of sparkling stars and nebulae. Marked a tiny red marker pin on the Orion Arm: 'You Are Here!'

- **Phase 2: Building the 3D Solar System Model:**

- Rolled newspaper soaked in flour glue into different sized spheres for the 8 planets and the central Sun.
- Painted each planet in authentic astronomical colors: fiery yellow-orange for Sun, blue-green for Earth, rusty red for Mars, striped brown for giant Jupiter, and added cardboard rings to Saturn.
- Mounted the planets on wooden skewers arranged in correct sequence from the Sun.



Planetary Characteristics & Model Log Table:-

The table below logs the planets in order from the Sun, their craft representations, and key facts:

<i>Celestial Body</i>	<i>Model Size & Color</i>	<i>Distance from Sun</i>	<i>Key Astronomical Feature</i>
The Sun	Largest sphere; bright yellow-orange	Center of Solar System	Yellow dwarf star powering all planetary orbits
Mercury	Tiny grey ball (1.5 cm)	58 million km	Smallest planet; closest to Sun; cratered surface
Venus	Yellowish-white sphere (3.0 cm)	108 million km	Hottest planet; thick runaway greenhouse atmosphere
Earth	Blue-green-white marble (3.2 cm)	150 million km	Our home planet; liquid water and living biosphere
Mars	Rusty reddish-orange ball (2.2 cm)	228 million km	The Red Planet; iron oxide soil; polar ice caps
Jupiter	Giant striped ball (7.0 cm)	778 million km	Largest planet; gas giant; famous Great Red Spot storm
Saturn	Golden sphere with wide cardboard rings	1,427 million km	Spectacular icy ring system; lowest planetary density
Uranus	Pale cyan blue ball (4.0 cm)	2,871 million km	Ice giant; rotates completely on its side!
Neptune	Deep azure blue ball (3.8 cm)	4,498 million km	Farthest gas giant; extreme supersonic winds

Scientific Analysis & Cosmic Perspectives:-

Constructing these models provided profound scientific insights into cosmic architecture:

- **The Scale of Space:** If the Sun were scaled down to the size of a standard basketball, the Earth would be a tiny mustard seed 26 meters away, and Neptune would be a peanut 800 meters down the street! Space is predominantly empty void held together by gravitational fields.
- **The Sun's Dominant Mass:** The Sun contains 99.86% of all total mass in the entire Solar System. Its immense gravitational pull keeps all planets, asteroids, and comets bound in fixed elliptical orbits.
- **Galactic Orbit of the Solar System:** Just as planets revolve around the Sun, our entire Solar System hurtles through space at an astonishing speed of 828,000 km/h, taking approximately **230 million years** to complete one full revolution around the Milky Way galactic center!

Everyday Applications & Space Exploration:-

Studying our Solar System and Milky Way drives human technological innovation:

- **Satellite Technology:** Understanding planetary orbits enabled scientists to launch weather satellites, GPS navigation, and communication satellites that power our modern internet and smartphones.
- **India's Space Achievements (ISRO):**
 - *Chandrayaan-3 (2023):* India became the first nation in history to land a robotic rover near the lunar South Pole!
 - *Mangalyaan (Mars Orbiter Mission):* Successfully orbited Mars on India's maiden attempt.
 - *Aditya-L1 (2023):* India's solar observatory studying solar flares at the Sun-Earth L1 point.
- **Search for Extraterrestrial Life:** Astronomers study Mars and icy moons (Europa and Enceladus) to search for microbial life beneath ice sheets.

What We Learned & Conclusion:-

- Our Solar System is a tiny family of eight planets orbiting the Sun inside the giant Milky Way galaxy.
- The Milky Way is a barred spiral galaxy containing hundreds of billions of stars spanning 1,00,000 light-years.
- Planets orbit in elliptical paths maintained by the Sun's immense gravitational anchor.
- Making 3D models with papier-mâché and cotton transforms abstract astronomy into visible, hands-on learning.

Conclusion:-

Building models of the Milky Way galaxy and the Solar System was an exhilarating creative and scientific journey. Using simple art materials like cardboard, papier-mâché, cotton, and glitter, our class visualized the grand architecture of our cosmos clearly. The project fostered deep curiosity about space exploration, astronomy, and our planet's humble, precious place in the vast cosmic universe.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 19: Life Cycle of Stars).
- ISRO Space Science Guide for Students: Solar System & Planetary Missions.
- National Geographic Visual Atlas of the Universe for Young Astronomers.